

Linear Models with a factor as predictor variable Analysis of variance - ANOVA

1

Analysis of variance - ANOVA

An ANOVA is a Linear model with one or more discrete predictor variables (i.e. factor(s) with 2 or more levels)

$$y_{ik} = a_i x_{ik} + \varepsilon_{ik}$$

Response variable Predictor variable(s) Residuals

There are i levels in the predictor variable, and k individuals in each of the i levels.

2

ANOVA

Testing the effect of two different fertilizers on plant growth

3 treatments:

- Control (soil without additional fertilizer)
- Fertilizer 1 (soil + fertilizer 1)
- Fertilizer 2 (soil + fertilizer 2)

50 plants per treatment

Growth rate measured has mm per day



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3

3

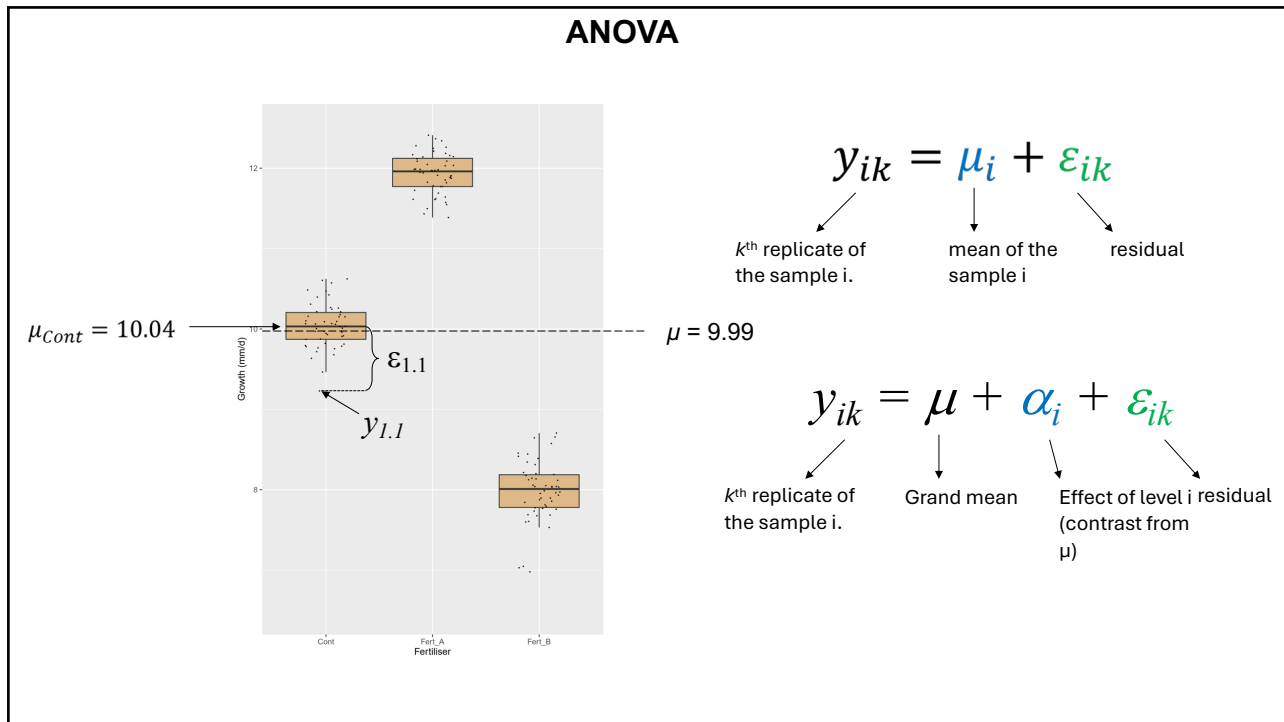
ANOVA

$$y_{ik} = a_i x_{ik} + \varepsilon_{ik}$$

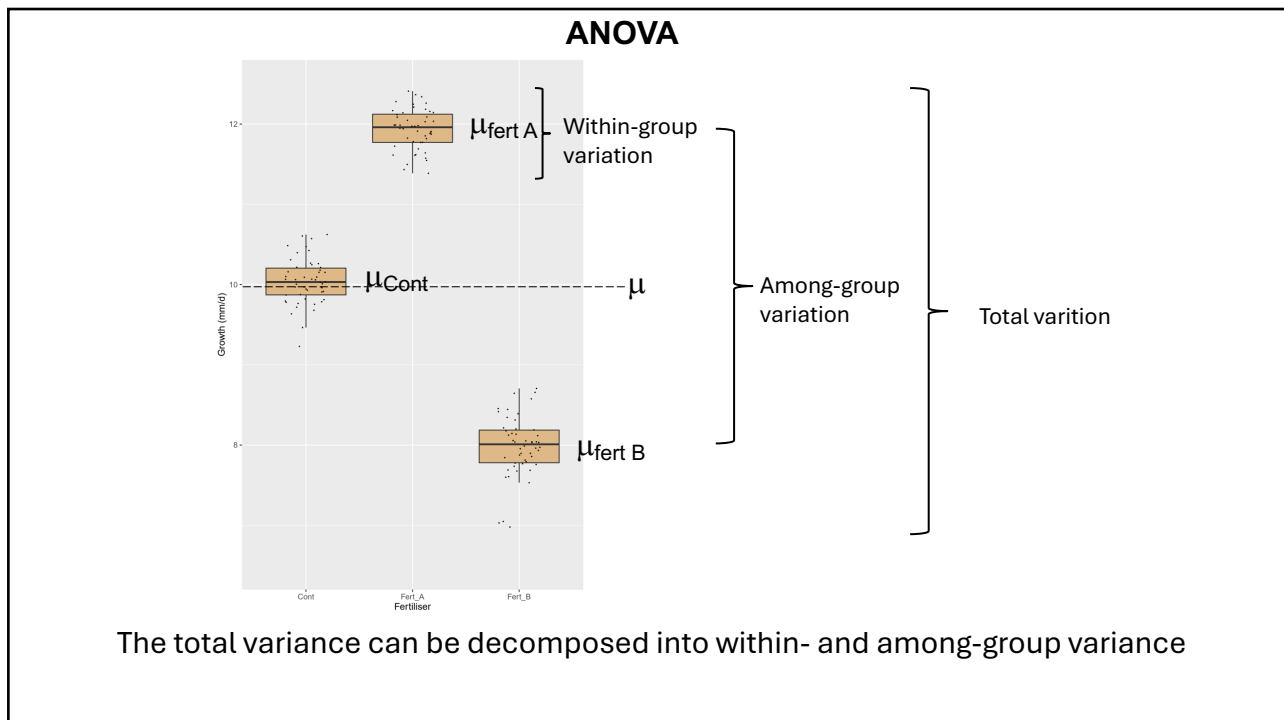
Plant growth mm.d⁻¹ Fertilization treatment (Cont, F-1, F-2) Residuals

There are 3 levels in the predictor variable ($i = 3$), and 50 observations per levels ($k = 50$).

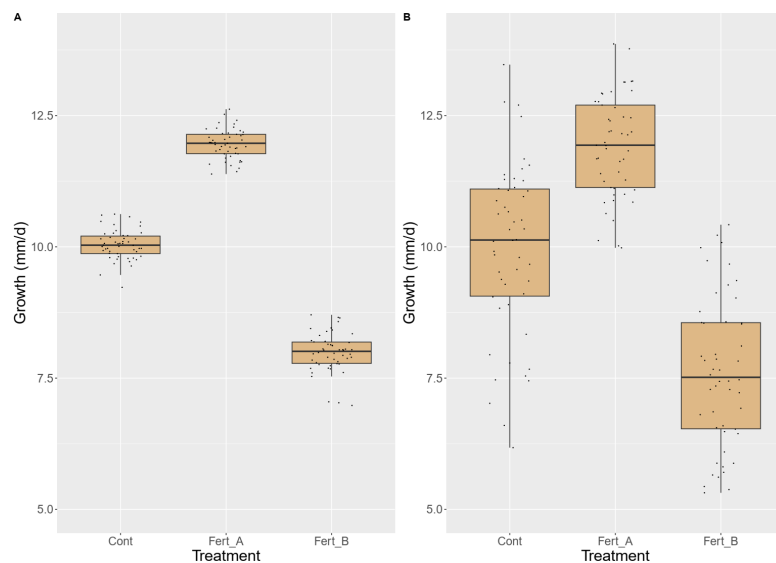
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5



6



Strong variation due to the treatment compared to the residual variation

Low variation due to the treatment compared to the residual variation

7

Statistical hypotheses

$$y_{ik} = \mu_i + \varepsilon_{ik}$$

$\swarrow$ $\downarrow$ $\searrow$
 k^{th} replicate of mean of the residual
the sample i . sample i

Null hypothesis: $H_0 = \mu_1 = \mu_2 = \mu_3 = \mu$

Alternative hypothesis: $H_1 = \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu$

The model compares the residual variance under both hypotheses. To do this, it calculates the relative importance of the variance explained by the treatment (**among-group**) with the residual variance (**within-group**).

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8

8

Calculating the different components of the variance

Different components of the variance

We need to calculate 3 sum-of-square residuals

$$SS_{among} = n \sum^a (\bar{Y} - \bar{\bar{Y}})^2$$

$$SS_{within} = \sum^a \sum^n (Y - \bar{Y})^2$$

$$SS_{total} = \sum^a \sum^n (Y - \bar{\bar{Y}})^2$$

The variance is the average square residual, that is, the SSR divided by the degrees of freedom ($\approx$ number of observations)

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9

9

The ANOVA Table

<i>Source of variation</i>	<i>df</i>	<i>SS</i>	<i>MS</i>
Among groups	$a - 1$	$n \sum^a (\bar{Y} - \bar{\bar{Y}})^2$	$SS_a / (a-1)$
Within groups (error)	$a(n-1)$	$\sum^a \sum^n (Y - \bar{Y})^2$	$SS_w / a(n-1)$
Total	$an - 1$	$\sum^a \sum^n (Y - \bar{\bar{Y}})^2$	

a = number of categories

n = number of observations in each category

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10

10

The ANOVA Table

Source of variation	df	SS	MS	F
Among groups	a - 1	$n \sum (\bar{y} - \bar{\bar{y}})^2$	$SS_a / (a-1)$	$\frac{SS_a / (a-1)}{SS_w / a(n-1)}$
Within groups	a(n-1)	$\sum \sum (Y - \bar{Y})^2$	$SS_w / a(n-1)$	
Total	an - 1	$\sum \sum (Y - \bar{\bar{Y}})^2$		

Test Statistic = $MS_{\text{model}} / MS_{\text{Res}}$; under H_0 it follows a F distribution with a-1 and N-a d.f.

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11

11

Let's see an example

cat 1	cat 2	cat3
2	3	5
3	4	6
3	3	5
2	4	6
3	3	5

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12

12

ANOVA Table

<i>Source of variation</i>	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>
Among groups				
Within groups <i>error</i>				
Total				

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13

13

Fitting the model in 
and interpreting the output

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14

14

Linear model with discrete predictor variable: Assumptions

Independence of the data within and among samples

Homogeneity of the variances

Normal distribution of the residuals

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15

15

Non-independence of the data

Positive "correlation" within treatment:

the "behaviour" of one experimental unit within one treatment positively affect (regarding the recorded variable) the "behaviour" of the other experimental units.

This may affect:

-the mean of the recorded variable

-Decrease the within sample variance



Excess in Type I error

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16

16

Non-independence of the data

Negative "correlation" within treatment:

the "behaviour" of one experimental unit within one treatment negatively affect (regarding the recorded variable) the "behaviour" of the other experimental units.

This may affect:

- the mean of the recorded variable
- increase the within sample variance

 Excess in Type II error

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17

17

Non-independence of the data: summary

Non-independence within treatments	Non-independence among treatments
<i>Positive correlation</i>	
σ_e^2 Within sample underestimated	σ_e^2 Within sample overestimated
F-ratio excessive	F-ratio too small
Increase type I error	Increase type II error
Spurious difference detected	Real differences not detected
<i>Negative correlation</i>	
σ_e^2 Within sample overestimated	σ_e^2 Within sample underestimated
F-ratio too small	F-ratio excessive
Increase type II error	Increase type I error
Real differences not detected	Spurious difference detected

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18

18

Homogeneity of variance

Samples from populations with different variances but the same mean can differ in more ways than samples from populations with the same variances

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19

19

Linear model with a factor as predictor variable

Exercise 1 on yield and fertilizer

Data set: fertilizer (From Grafen and Hails 2004)

These data present the yield (tonnes per ha.), for a crop under three different fertilization treatments. You will test whether the yield is affected by the treatment.

- Analyse the data using a linear model. You will particularly pay attention to the diagnostic of the model and the possible effect of outliers.
- Write a rapid presentation of the results including the necessary figure(s), summary statistics and effect size. You can also include some text about the effect of the outliers.

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20

20

Independence of data - a thought exercise

Hypothesis: the intensity of the begging behavior by chicks is related to their state of hunger.

Prediction: chicks that have been starved for a short period beg more than chicks that have been normally fed.

Experiment:

- For a random selection of nests, feeding parents are capture for a short period of time to prevent them feeding their chicks.
- Begging behaviour of each chick toward a dummy parent is recorded (video) for the starved chicks and for chicks that have been fed normally during the same period (control).

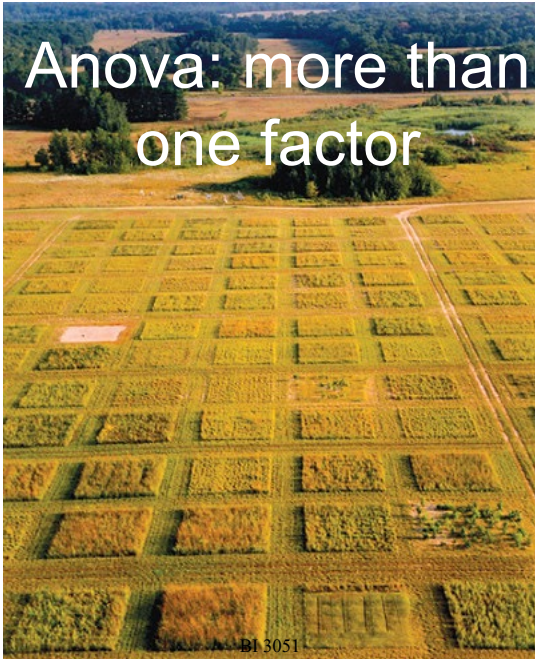
Is there problems of non-independence of the data?

How can we account for these when comparing the two groups?

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21

21



Anova: more than
one factor

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22

22

Full factorial design: Crossed with same number of replicates

		Factor B with q levels			
		B ₁	B ₂	...	B _q
Factor A with p levels	A ₁	n	n	...	n
	A ₂	n	n	...	n
	...				
	A _p	n	n	...	n

Number of observations $N = npq$

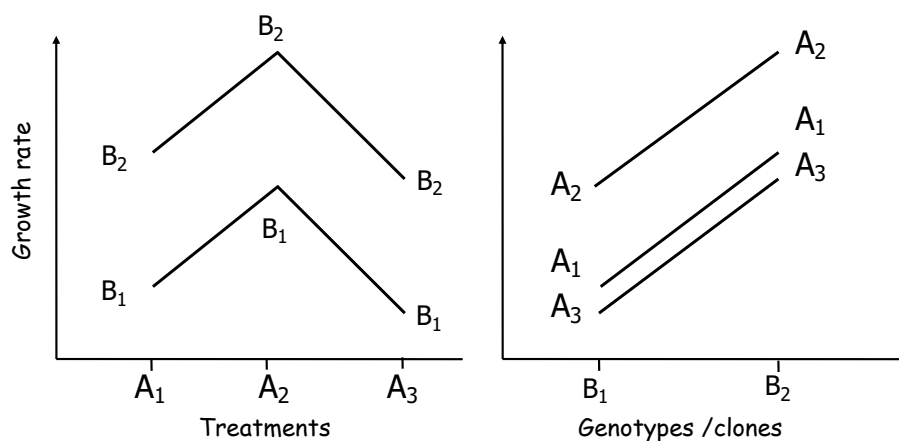
Number of experimental units (not replicated) = pq

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23

23

Additive effects:
2 genotypes in 3 environments

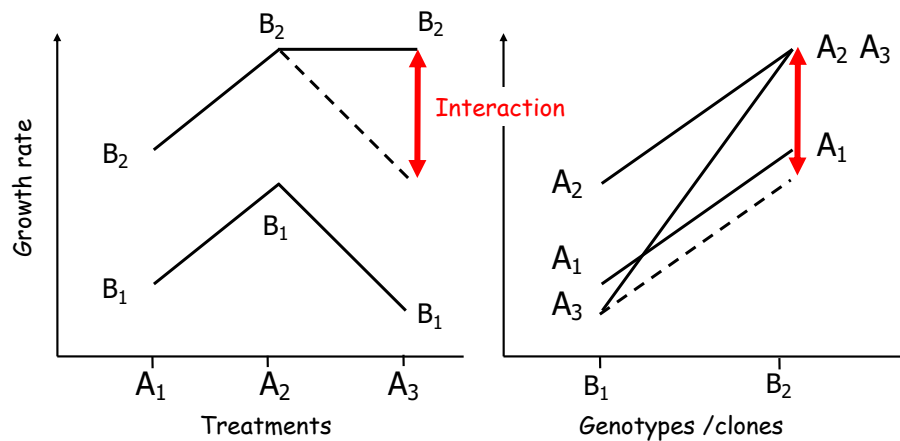


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24

24

Interaction

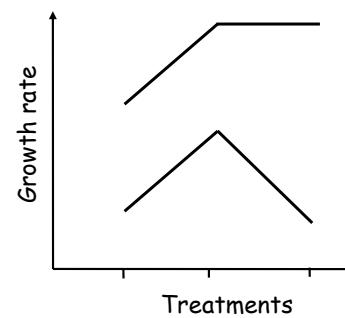


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25

25

Interaction



$$X_{ijk} = \mu + A_i + B_j + AB_{ij} + \varepsilon(AB)_{k(ij)}$$

μ : grand mean

A_i : level i for the factor A which comprises a levels

B_j : level j for the factor B which comprises b levels

AB_{ij} : interaction term

$\varepsilon(AB)_{k(ij)}$: individual residuals (error term)

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26

26

Decomposing the variance

$\varepsilon (AB)_{k(ij)} = X_{ijk} - \mu_{ij}$ is the residual.

σ_e^2 the residual variance – assumed to be equal for all $a*b$ populations.

μ is the mean of all $a*b$ μ_{ij} .

$A_i = \mu_i - \mu$ (**Main effects of factor A**) where μ_i is the mean when factor A = i

$B_j = \mu_j - \mu$ (**Main effects of factor B**) where μ_j is the mean when factor B = j

$AB_{ij} = \mu_{ij} - \mu - A_i - B_j = \mu_{ij} - \mu_i - \mu_j + \mu$ is the **interaction effect** for the combination of level i for factor A and level j for factor B . (AB_{ij} is equal to zero if $\mu_{ij} = \mu + A_i + B_j$. i.e. additive effect only)

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27

27

Decomposing the variance and tests

Source of variation	Sum of squares	Degrees of freedom	Mean square estimates	F-ratio versus
Among A	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_i - \bar{X})^2$	$a - 1$	$\sigma_e^2 + bnk_A^2$	Residuals
Among B	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_j - \bar{X})^2$	$b - 1$	$\sigma_e^2 + ank_B^2$	Residuals
A $\times$ B	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_{ij} - \bar{X}_i - \bar{X}_j + \bar{X})^2$	$(a - 1)(b - 1)$	$\sigma_e^2 + nk_{AB}^2$	Residuals
Residuals	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_{ijk} - \bar{X}_{ij})^2$	$ab(n - 1)$	σ_e^2	
Total	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (X_{ijk} - \bar{X})^2$	$abn - 1$	–	

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28

28

Decomposing the variance

Source of variation	Sum of squares	Degrees of freedom	Mean square estimates
Among A	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_i - \bar{X})^2$	$a - 1$	$\sigma_e^2 + bnk_A^2$
Among B	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_j - \bar{X})^2$	$b - 1$	$\sigma_e^2 + ank_B^2$
A × B	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_{ij} - \bar{X}_i - \bar{X}_j + \bar{X})^2$	$(a - 1)(b - 1)$	$\sigma_e^2 + nk_{AB}^2$
Residuals	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (\bar{X}_{ijk} - \bar{X}_{ij})^2$	$ab(n - 1)$	σ_e^2
Total	$\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (X_{ijk} - \bar{X})^2$	$abn - 1$	—

$$k_A^2 = \frac{\sum_{i=1}^a (A_i - \bar{A})^2}{(a-1)} = \frac{\sum_{i=1}^a A_i^2}{(a-1)}, \text{ because } \sum_{i=1}^a A_i = 0$$

$$k_B^2 = \frac{\sum_{j=1}^b (B_j - \bar{B})^2}{(b-1)} = \frac{\sum_{j=1}^b B_j^2}{(b-1)}, \text{ because } \sum_{j=1}^b B_j = 0$$

$$k_{AB}^2 = \frac{\sum_{i=1}^a \sum_{j=1}^b (AB_{ij} - \bar{A}\bar{B} - (\bar{A}_i - \bar{A})(\bar{B}_j - \bar{B}))^2}{(a-1)(b-1)} = \frac{\sum_{i=1}^a \sum_{j=1}^b AB_{ij}^2}{(a-1)(b-1)}$$

because $\sum_{j=1}^b AB_{ij} = 0$ for all i and $\sum_{i=1}^a AB_{ij} = 0$ for all j

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29

29

Linear model with multiple factors as predictor variables

Exercise 2: Testing the effect of water and brand of detergent on different clones of Daphnia

Data set: daphnia (from the web site accompanying Crawley's book)

The data represent the growth rate of different clones of Daphnia in different rivers when they are exposed to different types of detergent.

Analyse these data and present rapidly the results in a written format with the necessary statistics and graphs. Notice that you can break down the analysis into several “sub-analyses” to understand the exact effects of the detergents and the water on the growth of Daphnia.

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30

30

Linear model with multiple factors as predictor variables

Biological question: Is antler size in roe deer affected by the population density?

Data: Left and right antler length measured on roe deer culled in a population south west of Amsterdam (Netherlands). Each year, transects were performed to estimate the population density and roe deer were shot to limit the population growth.

The population density has been categorized as: low, intermediate and high.

We recognise three age categories: yearling (1 year old), subadult (2 year old), adult (3 years and older).

Fit a simple linear model in R where you test whether population density affect the antler size.

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31